

- is obese (BMI >30)
- has pre-diabetes (fasting plasma glucose 100–125 mg/dL)
- has diabetes (fasting plasma glucose >126 mg/dL)
- has hypertension (BP >140/90 mmHg)
- has dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol)
- Treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management

Monitoring after starting an atypical antipsychotic

- * BMI monthly for 3 months, then quarterly
- * Consider monitoring fasting triglycerides monthly for several months in patients at high risk for metabolic complications and when initiating or switching antipsychotics
- * Blood pressure, fasting plasma glucose, fasting lipids within 3 months and then annually, but earlier and more frequently for patients with diabetes or who have gained >5% of initial weight
- Treat or refer for treatment and consider switching to another atypical antipsychotic for patients who become overweight, obese, pre-diabetic, diabetic, hypertensive, or dyslipidemic while receiving an atypical antipsychotic

* Even in patients without known diabetes, be vigilant for the rare but life-threatening onset of diabetic ketoacidosis, which always requires immediate treatment, by monitoring for the rapid onset of polyuria, polydipsia, weight loss, nausea, vomiting, dehydration, rapid respiration, weakness and clouding of sensorium, even coma

- Patients with low white blood cell count (WBC) or history of drug-induced leukopenia/neutropenia should have complete blood count (CBC) monitored frequently during the first few months and asenapine should be discontinued at the first sign of decline in WBC in the absence of other causative factors
- Patients should be monitored periodically for the development of abnormal movements of tardive dyskinesia, using neurological exam and the AIMS (Abnormal Involuntary Movement Scale)

Side Effects

How Drug Causes Side Effects

- Acute blockade of dopamine 2 receptors in the striatum can cause drug-induced parkinsonism, dystonia, or akathisia
- Chronic blockade of dopamine 2 receptors in the striatum can cause tardive dyskinesia
- By blocking dopamine 2 receptors in the pituitary, it can cause elevations in prolactin

- Blocking alpha 1 adrenergic receptors can cause dizziness, hypotension, and syncope
- Mechanism of weight gain and increased incidence of diabetes and dyslipidemia with atypical antipsychotics is unknown, and risks vary based on the particular agent

Notable Side Effects

- * Sedation, dizziness
- Oral hypoesthesia
- Application site reactions (oral ulcers, blisters, peeling/sloughing, inflammation for sublingual; irritation, burning for transdermal)
- * Drug-induced parkinsonism, akathisia
- * May increase risk for diabetes and dyslipidemia
- Tardive dyskinesia (reduced risk compared to conventional antipsychotics)
- Risk of potentially irreversible involuntary dyskinetic movements may increase with cumulative dose and treatment duration
- Orthostatic hypotension

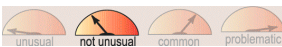


Life-Threatening or Dangerous Side Effects

- Type 1 hypersensitivity reactions (anaphylaxis, angioedema, low blood pressure, rapid heart rate, swollen tongue, difficulty breathing, wheezing, rash)

- Hyperglycemia, in some cases extreme and associated with ketoacidosis or hyperosmolar coma or death, has been reported in patients taking atypical antipsychotics
- As a class, antipsychotics are associated with an increased risk of death and cerebrovascular events in elderly patients with dementia; not approved for treatment of dementia-related psychosis
- Rare neuroleptic malignant syndrome (NMS) may cause hyperpyrexia, muscle rigidity, delirium, and autonomic instability with elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure
- Rare seizures

Weight Gain



- Occurs in a significant minority
- May be less than for some antipsychotics, more than for others

Sedation



- Many experience and/or can be significant in amount

What to Do About Side Effects

- Wait
- Wait

- Wait
- Anticholinergics may reduce drug-induced parkinsonism when present
- Beta blockers, benzodiazepines, or 5HT2A antagonists (e.g., mirtazapine, cyproheptadine) may reduce akathisia
- Weight loss, exercise programs, and medical management for high BMIs, diabetes, dyslipidemia
- Metformin may help prevent or reverse antipsychotic-induced weight gain
- Switch to another atypical antipsychotic

Best Augmenting Agents for Side Effects

- Beta blockers, benzodiazepines, or 5HT2A antagonists (e.g., mirtazapine, cyproheptadine) for akathisia
- Benztropine, trihexyphenidyl, or amantadine for drug-induced parkinsonism
- Valbenazine or deutetrabenazine for tardive dyskinesia
- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- Schizophrenia and bipolar mania (sublingual): 10–20 mg/day in 2 divided doses
- Schizophrenia (transdermal): 3.8 mg/24 hours

Dosage Forms

- Sublingual tablet 2.5 mg, 5 mg, 10 mg
- Transdermal system 3.8 mg/24 hours, 5.7 mg/24 hours, 7.6 mg/24 hours

How to Dose –Sublingual

- Must be administered sublingually; patients may not eat or drink for 10 minutes following administration
- Schizophrenia: initial 10 mg/day in 2 divided doses; maximum dose generally 20 mg/day in 2 divided doses; limited experience with once daily administration
- Bipolar mania (adults, monotherapy): initial 20 mg/day in 2 divided doses; can reduce dose to 10 mg/day in 2 divided doses if there are adverse effects
- Bipolar mania (adults, adjunct): initial 10 mg/day in 2 divided doses; can increase to 20 mg/day in 2 divided doses
- Bipolar mania (children, monotherapy): initial 5 mg/day in 2 divided doses; after 3 days can increase to 10 mg/day in 2 divided doses; after 3 more days can increase to 20 mg/day in 2 divided doses

- Pediatric patients may be more sensitive to dystonia with initial dosing if the recommended titration schedule is not followed

How to Dose –Transdermal

- Apply one Secuado transdermal system every 24 hours to one of the following sites: hip, abdomen, upper arm, or lower back area
- Recommended starting dose: 3.8 mg/24 hours
- After 1 week, may increase to 5.7 mg/24 hours or 7.6 mg/24 hours



Dosing Tips –Sublingual

- Asenapine is not absorbed after swallowing (less than 2% bioavailable orally) and thus must be administered sublingually (35% bioavailable), as swallowing would render asenapine inactive
- Patients should be instructed to place the tablet under the tongue and allow it to dissolve completely, which will occur in seconds; tablet should not be divided, crushed, chewed, or swallowed
- Patients may not eat or drink for 10 minutes following sublingual administration so that the drug in the oral cavity can be absorbed locally and not washed into the stomach (where it would not be absorbed)
- Once daily use seems theoretically possible because the half-life of asenapine is 13–39 hours, but this has not been extensively studied and may be limited by the need to expose the limited sublingual surface area to a limited amount of sublingual drug dosage

- Some patients may respond to doses greater than 20 mg/day but no single administration should be greater than 10 mg, thus necessitating 3 or 4 separate daily doses
- Rather than raise the dose above normal dosing in acutely agitated patients requiring acute antipsychotic actions, consider short-term (one dose or more) augmentation with a benzodiazepine or another antipsychotic, especially intramuscularly
- Rather than raise the dose above normal dosing in partial responders, consider augmentation with a mood-stabilizing anticonvulsant, such as valproate, topiramate, or lamotrigine
- Children and elderly should generally be dosed at the lower end of the dosage spectrum
- Due to rapid onset of action, can be used as a rapid acting “prn” or “as needed” dose for agitation or transient worsening of psychosis or mania instead of an injection
- Treatment should be suspended if absolute neutrophil count falls below 1000/mm³

Dosing Tips –Transdermal

- Based on the average exposure (AUC) of asenapine, Secuado 3.8 mg/24 hours corresponds to 5 mg twice daily of sublingual asenapine, and Secuado 7.6 mg/24 hours corresponds to 10 mg twice daily of sublingual asenapine

- Instruct patient to select a different transdermal system application site each day to limit the occurrence of skin reactions
- Do not cut Secuado; the whole transdermal system should be applied
- Discard Secuado by folding the used transdermal system so that the adhesive side sticks to itself, and dispose of safely
- If irritation or a burning sensation occurs while wearing Secuado, remove the system and apply a new transdermal system to a new application site
- Showering is permitted, but the use of Secuado during swimming or taking a bath has not been evaluated
- Treatment should be suspended if absolute neutrophil count falls below $1000/\text{mm}^3$

Overdose

- Agitation, confusion

Long-Term Use

- Should periodically reevaluate long-term usefulness in individual patients, but treatment may need to continue for many years

Habit Forming

- No

How to Stop

- Down-titration, over 2–4 weeks when possible, especially when simultaneously beginning a new antipsychotic while switching (i.e., cross-titration)
- Rapid discontinuation could theoretically lead to rebound psychosis and worsening of symptoms

Pharmacokinetics

- Half-life 13–39 hours
- Inhibits CYP450 2D6
- Substrate for CYP450 1A2
- Optimal bioavailability is with sublingual administration (~35%); if food or liquid is consumed within 10 minutes of administration bioavailability decreases to 28%; bioavailability decreases to 2% if swallowed
- Transdermal: maximum asenapine concentrations are typically reached between 12 and 24 hours, with sustained concentrations during wear time (24 hours); following removal, elimination half-life is approximately 30 hours



Drug Interactions

- May increase effects of antihypertensive agents
- May antagonize levodopa, dopamine agonists

- CYP450 1A2 inhibitors (e.g., fluvoxamine) can raise asenapine levels
- Via CYP450 2D6 inhibition, asenapine could theoretically interfere with the analgesic effects of codeine, and increase the plasma levels of some beta blockers and of atomoxetine
- Via CYP450 2D6 inhibition, asenapine could theoretically increase concentrations of thioridazine and cause dangerous cardiac arrhythmias

Other Warnings/Precautions

- Use with caution in patients with conditions that predispose to hypotension (dehydration, overheating)
- Atypical antipsychotics have the potential for cognitive and motor impairment, especially related to sedation
- Atypical antipsychotics can cause body temperature dysregulation; use caution in patients who may experience conditions that increase body temperature (e.g., strenuous exercise, saunas, extreme heat, dehydration, or concomitant anticholinergic medication)
- Dysphagia has been associated with antipsychotic use, and asenapine should be used cautiously in patients at risk for aspiration pneumonia
- As with any antipsychotic, use with caution in patients with history of seizures

- Antipsychotics may cause somnolence, postural hypotension, motor and sensory instability, which may lead to falls; for patients with diseases, conditions, or medications that could exacerbate these effects, complete fall risk assessments when initiating antipsychotic treatment, and recurrently for patients on long-term antipsychotic therapy
- Avoid exposing transdermal system to external heat sources during wear, because both the rate and extent of absorption are increased
- During wear time or immediately after removal of transdermal system, local skin reactions may occur

Do Not Use

- Severe hepatic impairment (Child-Pugh C)
- If there is a proven allergy to asenapine or any components in the transdermal system

Special Populations

Renal Impairment

- Dose adjustment not generally necessary

Hepatic Impairment

- No dose adjustment necessary for mild to moderate impairment
- Contraindicated in patients with severe hepatic impairment

Cardiac Impairment

- Use in patients with cardiac impairment has not been studied, so use with caution because of risk of orthostatic hypotension
- Use with caution if patient is taking concomitant antihypertensive or alpha 1 antagonist

Elderly

- Some patients may tolerate lower doses better
- Although atypical antipsychotics are commonly used for behavioral disturbances in dementia, no agent has been approved for treatment of elderly patients with behavioral symptoms of dementia such as agitation
- Elderly patients with dementia-related psychosis treated with atypical antipsychotics are at an increased risk of death compared to placebo, and also have an increased risk of cerebrovascular events
- Consider pimavanserin for dementia-related psychosis or Parkinson's disease psychosis instead of an atypical antipsychotic



Children and Adolescents

- Transdermal asenapine is approved only in adults
- Sublingual asenapine is approved to treat acute manic/mixed episodes of bipolar I disorder in children ages 10 and older
- Efficacy of sublingual asenapine for schizophrenia was not demonstrated in an 8-week, placebo-controlled, double-blind trial in

adolescent patients ages 12 to 17 years

- Children and adolescents using asenapine may need to be monitored more often than adults and may tolerate lower doses better

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- There is a risk of abnormal muscle movements and withdrawal symptoms in newborns whose mothers took an antipsychotic during the third trimester; symptoms may include agitation, abnormally increased or decreased muscle tone, tremor, sleepiness, severe difficulty breathing, and difficulty feeding
- In animal studies, asenapine increased post-implantation loss and decreased pup weight and survival at doses similar to or less than recommended clinical doses; there was no increase in the incidence of structural abnormalities
- Psychotic symptoms may worsen during pregnancy and some form of treatment may be necessary

- National Pregnancy Registry for Atypical Antipsychotics: 1-866-961-2388,
<https://womensmentalhealth.org/research/pregnancyregistry/atypical/antipsychotic/>

Breast Feeding

- Unknown if asenapine is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed
- Infants of women who choose to breast feed while on asenapine should be monitored for possible adverse effects

The Art of Psychopharmacology

Potential Advantages

- Patients requiring rapid onset of antipsychotic action without dosage titration

Potential Disadvantages

- Patients who are less likely to be adherent
- Efficacy for schizophrenia was not demonstrated in an 8-week, placebo-controlled, double-blind trial in adolescent patients ages 12 to 17 years

Primary Target Symptoms

- Positive symptoms of psychosis
- Negative symptoms of psychosis
- Cognitive symptoms
- Unstable mood (both depression and mania)
- Aggressive symptoms

Pearls

- Asenapine's chemical structure is related to the antidepressant mirtazapine and it shares many of the same pharmacologic binding properties of mirtazapine plus many others
- Not approved for depression, but binding properties suggest potential use in treatment-resistant and bipolar depression
- Sublingual administration may require prescribing asenapine to reliable, adherent patients, or those who have someone who can supervise drug administration
- Asenapine has a more rapid onset of action (C_{max}) when given sublingually than most atypical antipsychotics do when given orally, and thus may be suitable for use as a prn (as needed) agent to treat acute agitation in patients with schizophrenia when an injection is not possible
- Patients with inadequate responses to atypical antipsychotics may benefit from determination of plasma drug levels and, if low, a dosage increase even beyond the usual prescribing limits

- Patients with inadequate responses to atypical antipsychotics may also benefit from a trial of augmentation with a conventional antipsychotic or switching to a conventional antipsychotic
- However, long-term polypharmacy with a combination of a conventional antipsychotic with an atypical antipsychotic may combine their side effects without clearly augmenting the efficacy of either
- For treatment-resistant patients, especially those with impulsivity, aggression, violence, and self-harm, long-term polypharmacy with 2 atypical antipsychotics or with 1 atypical antipsychotic and 1 conventional antipsychotic may be useful or even necessary while closely monitoring
- In such cases, it may be beneficial to combine 1 depot antipsychotic with 1 oral antipsychotic
- Although a frequent practice by some prescribers, adding 2 conventional antipsychotics together has little rationale and may reduce tolerability without clearly enhancing efficacy

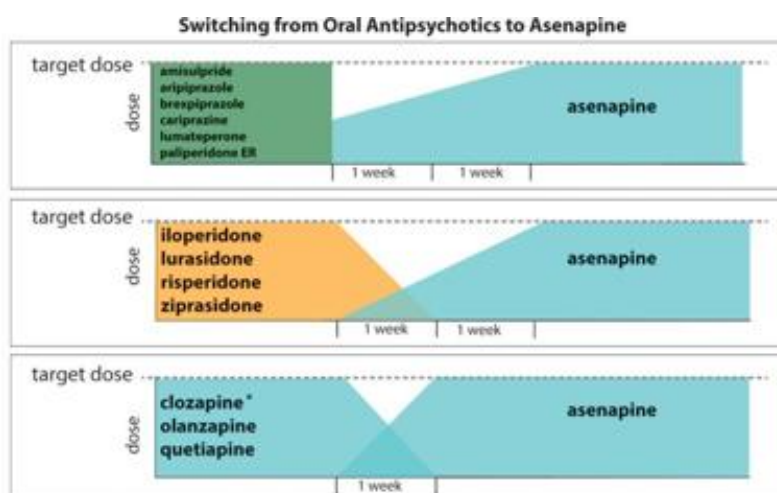
The Art of Switching



Switching from Oral Antipsychotics to Asenapine

- With amisulpride, aripiprazole, brexpiprazole, cariprazine, lurasidone, and paliperidone ER, immediate stop is possible; begin asenapine at middle dose

- With risperidone, ziprasidone, iloperidone, and lurasidone, begin asenapine gradually, titrating over at least 2 weeks to allow patients to become tolerant to the sedating effect
- * May need to taper clozapine slowly over 4 weeks or longer



Suggested Reading

Citrome L . Asenapine for schizophrenia and bipolar disorder: a review of the efficacy and safety profile for this newly approved sublingually absorbed second-generation antipsychotic . *Int J Clin Pract* 2009 ;**63** (12):1762–84 .

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Atomoxetine

Therapeutics

Brands

- Strattera
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: norepinephrine reuptake inhibitor (N-RI)
- Selective norepinephrine reuptake inhibitor (NRI)

Commonly Prescribed for

(bold for FDA approved)

- **Attention deficit hyperactivity disorder (ADHD) in adults and children over 6**
- Treatment-resistant depression

How the Drug Works

- Boosts neurotransmitter norepinephrine/noradrenaline and may also increase dopamine in prefrontal cortex
- Blocks norepinephrine reuptake pumps, also known as norepinephrine transporters
- Presumably this increases noradrenergic neurotransmission
- Since dopamine is inactivated by norepinephrine reuptake in frontal cortex, which largely lacks dopamine transporters, atomoxetine can also increase dopamine neurotransmission in this part of the brain

How Long Until It Works

- * Onset of therapeutic actions in ADHD can be seen as early as the first day of dosing
- Therapeutic actions may continue to improve for 8–12 weeks
- If it is not working within 6–8 weeks, it may not work at all

If It Works

- The goal of treatment of ADHD is reduction of symptoms of inattentiveness, motor hyperactivity, and/or impulsiveness that disrupt social, school, and/or occupational functioning
- Continue treatment until all symptoms are under control or improvement is stable and then continue treatment indefinitely as long as improvement persists
- Reevaluate the need for treatment periodically

- Treatment for ADHD begun in childhood may need to be continued into adolescence and adulthood if continued benefit is documented

If It Doesn't Work

- Consider adjusting dose or switching to another agent
- Consider behavioral therapy
- Consider the presence of noncompliance and counsel patient and parents
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., bipolar disorder, substance abuse, medical illness, etc.)
- Some patients may experience apparent lack of consistent efficacy due to activation of latent or underlying bipolar disorder, and require atomoxetine discontinuation and a switch to a mood stabilizer



Best Augmenting Combos for Partial Response or Treatment Resistance

- * Best to attempt other monotherapies prior to augmenting
- SSRIs, SNRIs, or mirtazapine for treatment-resistant depression (use combinations of antidepressants with atomoxetine with caution as this may theoretically activate bipolar disorder and suicidal ideation)
- Mood stabilizers or atypical antipsychotics for comorbid bipolar disorder

- For the expert, can combine with modafinil, methylphenidate, or amphetamine for ADHD

Tests

- None recommended for healthy patients
- May be prudent to monitor blood pressure and pulse when initiating treatment and until dosage increments have stabilized

Side Effects

How Drug Causes Side Effects

- Norepinephrine increases in parts of the brain and body and at receptors other than those that cause therapeutic actions (e.g., unwanted actions of norepinephrine on acetylcholine release causing decreased appetite, increased heart rate and blood pressure, dry mouth, urinary retention, etc.)
- Most side effects are immediate but often go away with time
- Lack of enhancing dopamine activity in limbic areas theoretically explains atomoxetine's lack of abuse potential

Notable Side Effects

- * Sedation, fatigue (particularly in children)
- * Decreased appetite
- * Rare priapism

- Increased heart rate (6–9 beats/min)
- Increased blood pressure (2–4 mmHg)
- Insomnia, dizziness, anxiety, agitation, aggression, irritability
- Dry mouth, constipation, nausea, vomiting, abdominal pain, dyspepsia
- Urinary hesitancy, urinary retention (older men)
- Dysmenorrhea, sweating
- Sexual dysfunction (men: decreased libido, erectile disturbance, impotence, ejaculatory dysfunction, abnormal orgasm; women: decreased libido, abnormal orgasm)



Life-Threatening or Dangerous Side Effects

- Increased heart rate and hypertension
- Orthostatic hypotension
- Severe liver damage (rare)
- Hypomania and, theoretically, rare induction of mania
- Rare activation of suicidal ideation and behavior (suicidality) (short-term studies did not show an increase in the risk of suicidality with antidepressants compared to placebo beyond age 24)

Weight Gain



- Reported but not expected

- Patients may experience weight loss

Sedation



- Occurs in significant minority, particularly in children

What to Do About Side Effects

- Wait
- Wait
- Wait
- Lower the dose
- If giving once daily, can change to split dose twice daily
- If atomoxetine is sedating, take at night to reduce daytime drowsiness
- In a few weeks, switch or add other drugs

Best Augmenting Agents for Side Effects

- For urinary hesitancy, give an alpha 1 blocker such as tamsulosin
- Often best to try another monotherapy prior to resorting to augmentation strategies to treat side effects
- Many side effects are dose-dependent (i.e., they increase as dose increases, or they reemerge until tolerance redevelops)